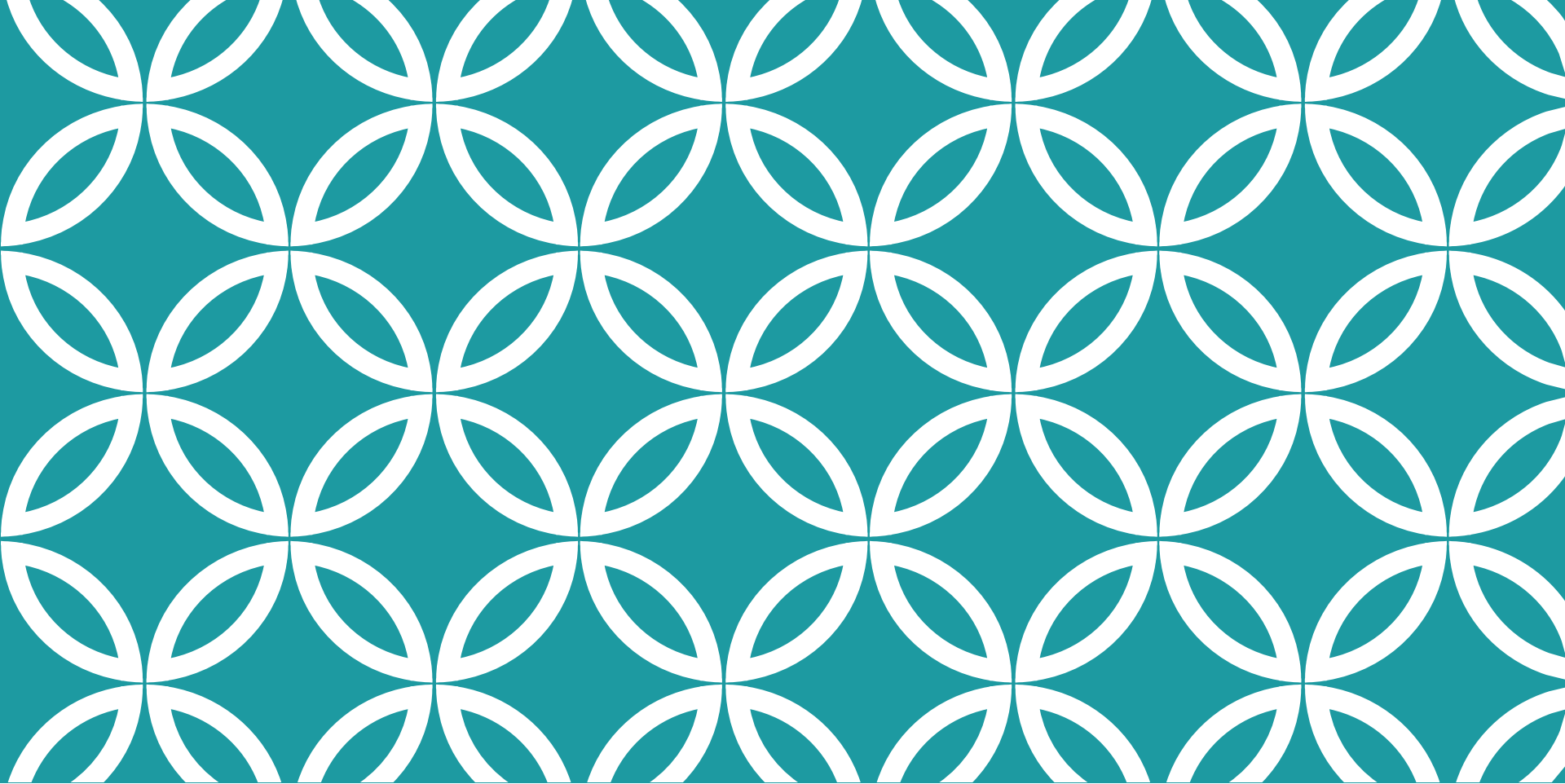


SPEECH AND VIDEO PROCESSING (CS30033)

PREPARED BY DR. JAYEETA CHAKRABORTY, ASSISTANT
PROFESSOR II, KIIT DU



COURSE INFORMATION

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COURSE OBJECTIVES

1. To introduce speech properties, issues, and speech-processing concepts for prediction
2. To learn feature extraction, selection, and speech recognition algorithms
3. To understand the video principles and its processing techniques
4. To learn motion estimation, optimization techniques, object tracking, boundary detection, compression techniques
5. To implement the algorithms using Python and auxiliary tools.

COURSE OUTCOMES

1. Assess the mechanisms of human speech production systems and methods for speech feature extraction
2. Apply the algorithms of speech analysis and speech recognition
3. Incorporate the fundamental techniques in digital video processing, including image characteristics and sensors
4. Implement motion estimation and object tracking algorithms on video sequence
5. Develop approximated real-life speech and video using Python
6. Apply suitable Learning Models in different problems to obtain desirable results

BOOKS

Text Books:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series.
2. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.
3. Digital Video processing, A Murat Tekalp, Prentice Hall.
4. Video Processing and Communications, Yao Wang, J. Osternann and Qin Zhang, Pearson Education.

Reference Books:

1. Speech and Audio Signal Processing, B. Gold and N. Morgan, Wiley.
2. Digital image sequence processing, Compression, and analysis, Todd R. Reed, CRC Press.
3. Handbook of Image and Video Processing, Al Bovik, Academic press, Second Edition.

GRADING POLICY

Assignments/Quizzes/ Tests/Activities: **30 Marks**

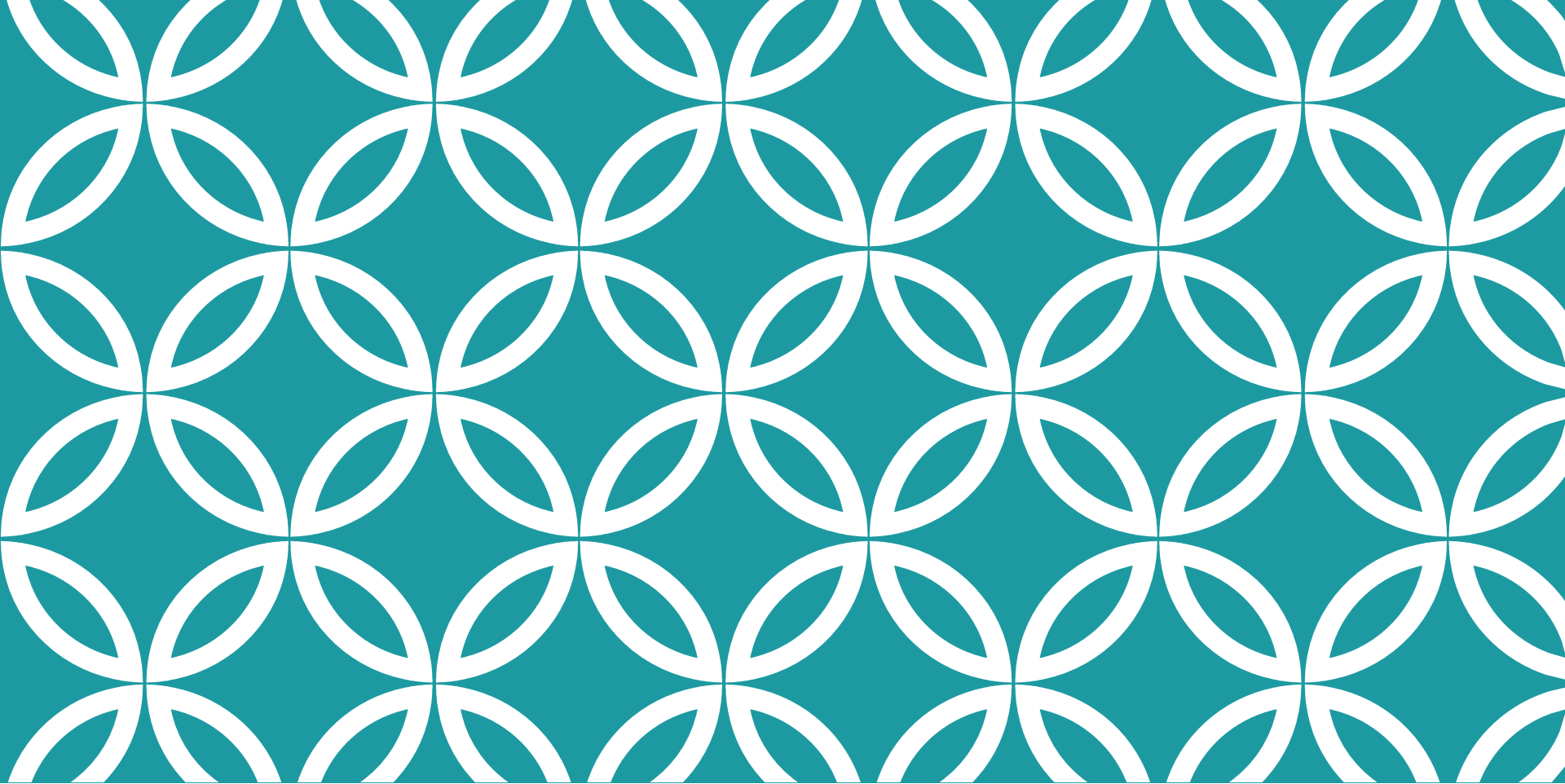
Mid-semester exam: **20 Marks**

End-semester exam: **50 Marks**

ACTIVITY SL.NO.	TYPE	DATE	MARKS	CO
1	Activity-1	20th July - 14th Aug	5	CO1, CO2
2	Activity-2	17th Aug - 31st Aug	5	CO1, CO2
3	Activity-3	01st Sep - 6th Sep	5	CO1, CO2, CO3
MID SEM EXAM 07th Sep-12th Sep 2026				
4	Activity-4	14th Sep – 30th Sep	5	CO3, CO4
5	Activity-5	1st Oct - 15th Oct	5	CO3, CO4
6	Activity-6	21st Oct - 6th Nov	5	CO3, CO4, CO5, CO6
END SEM EXAM 9th Nov - 18th Nov 2026				

*Tentative Activity
Calendar*

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DIGITAL SPEECH PROCESSING

Introduction
Speech Processing
Pipeline
Applications
Real-time Platforms

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DEFINITION

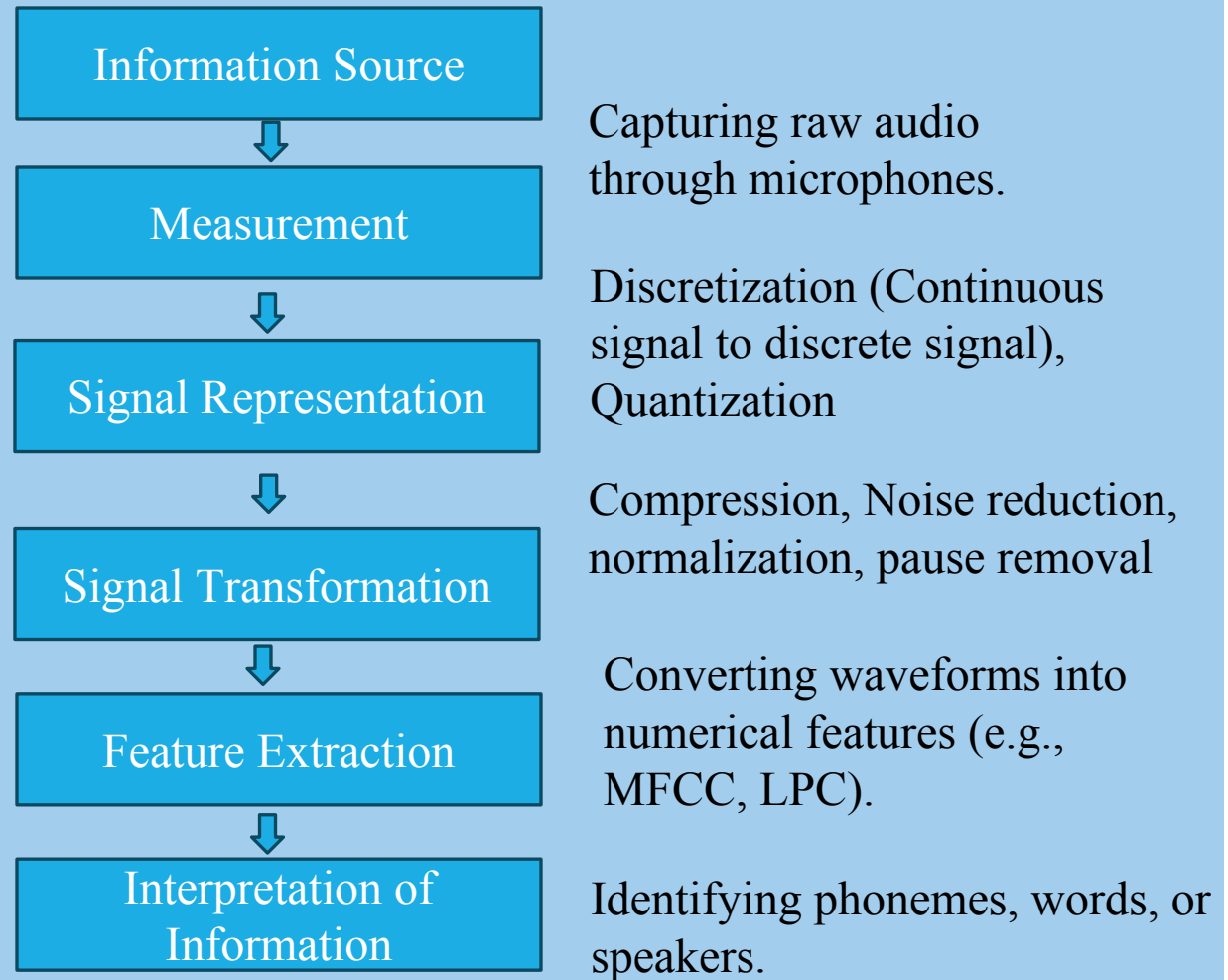
*“Speech processing is the study of speech signals and the processing methods of these signals. It involves **analysis, synthesis, recognition, and coding** of speech for applications in communication and human–computer interaction.”*

—from “Fundamentals of Speech Recognition” by Lawrence Rabiner & Biing-Hwang Juang (Prentice Hall, 1993)

*“Digital speech processing refers to **the representation, transformation, and interpretation** of speech signals using digital techniques, enabling compression, transmission, and recognition.”*

—from “Digital Processing of Speech Signals” by L. R. Rabiner & R. W. Schafer (Prentice Hall, 1978)

SPEECH PROCESSING MODEL/ PIPELINE



INTRODUCTION

Different disciplines that are involved in speech processing:

1. Signal Processing
2. Information Theory
3. Algorithms
4. Phonetics
5. Acoustics

SPEECH TRANSMISSION VS SPEECH PROCESSING

“Talking over a mobile phone is not the same as digital speech processing”

Mobile Phone Speech Transmission:

- ❑ **Analog/Digital Transmission** - Your voice is captured by the microphone, converted into electrical signals, and transmitted over cellular networks.
- ❑ **Compression & Encoding** - Uses codecs (like AMR, Opus) to compress audio for efficient bandwidth usage.
- ❑ **Network Transport** - Signals are carried over GSM, LTE, or VoIP protocols.
- ❑ **Goal:** Deliver your voice clearly to the other person, not analyze or interpret it.

Digital Speech Processing

- ❑ **Feature Extraction** - Converts raw audio into numerical features (MFCC, LPC).
- ❑ **Recognition & Interpretation** - Identifies phonemes, words, and meaning.
- ❑ **Applications** - Voice assistants, transcription, biometrics, translation.
- ❑ **Goal:** Understand, analyze, or act on speech content.

APPLICATIONS OF DIGITAL SPEECH PROCESSING

Smart Homes - Voice assistants (Alexa, Siri, Google Assistant) control appliances, lighting, and entertainment systems.

Education - Real-time captioning (transcription generation), pronunciation and fluency checks in language learning apps.

Healthcare – Doctor-patient conversation, prescription assistance, speech aid devices

Media – Live captioning for broadcasting and streaming platforms.

Business/Commerce – Automated customer service through speech-driven chatbots

Automotive - Voice commands for navigation, infotainment, and hands-free communication

Security – voice biometric authentication

REAL-TIME SPEECH PROCESSING PLATFORMS

Deepgram

- Known for **fastest English transcription** with ~250–300ms latency.
- Supports **speaker diarization, sentiment detection, and topic analysis**.
- Ideal for live apps like call centers, meetings, and customer support.

AssemblyAI

- Offers **live transcription APIs** with sub-300ms response.
- Features include **inline speaker labeling, real-time prompting, and unlimited concurrency**.
- Widely used in **voice agents and accessibility tools**.

OpenAI Realtime

- Integrates with GPT for **function calling and conversational AI**.
- Latency ~300–500ms, supports **99+ languages**.
- Great for **multilingual chatbots and interactive assistants**.

Google Cloud STT

- Supports **125+ languages** with auto punctuation.
- Strong enterprise compliance, though slightly higher latency (~400ms).
- Best for organizations already using Google Cloud ecosystem.